A Lightweight Convolutional Neural Network Based on Joint Correlation Distance Constraints and Density Peak Clustering for Hyperspectral Target Detection

Yanni Dong, *Senior Member, IEEE*, Xiuqing Dai, Yuxiang Zhang, and Bo Du, *Senior Member, IEEE*

***Abstract*— Deep learning can fully exploit the potential information of data, which facilitates effective target-background separation for hyperspectral target detection (HTD). The deep learning model usually requires a large number of labeled samples yet the available prior target spectra in the hyperspectral image (HSI) are extremely limited. In addition, network training also requires reliable input samples to ensure that the trained model has stronger data discrimination, but current detection methods often suffer from the problem that the background samples are impure. To address the above problems, we propose a lightweight convolutional neural network based on joint correlation distance constraints and density peak clustering (LCNN-CD) for HTD. First, the correlation distance between the prior target and HSI is calculated, and the pixels with high correlation are retained for expanding the target sample, which effectively handles the problem of insufficient target samples. Second, the density peak clustering algorithm is used to extract the main pure background samples, which effectively solves the problem that the background samples are sufficiently impure. Finally, a lightweight convolutional neural network model is designed, which is fed by the training samples (both target and background samples) to obtain the final detection results with low computational efficiency. The proposed method is compared with the classical and recently popular HTD methods on three real HSI datasets. Numerous experiments show that the proposed LCNN-CD method has better target detection performance and effectiveness.**

***Index Terms*— Correlation distance constraint, density peak clustering (DPC), hyperspectral target detection (HTD), lightweight convolutional neural network (CNN).**

I. Introduction

THE high spectral resolution property of hyperspectral image (HSI) enhances the ability to acquire diagnostic spectral feature information of materials, making it possible to distinguish different materials. Therefore, HSI has been used in various fields, such as unmixing [1], [2], dimensionality reduction [3], [4], classification [5], [6], band selection [7], [8], anomaly detection [9], [10], [11], and target detection [12], [13]. Among them, hyperspectral target detection (HTD) aims to detect and identify target pixels based on specific a prior spectral feature information, and it has high application value in military defense, mineral exploration, personnel search, and environmental monitoring [14], [15], [16], [17], [18], [19].

According to the target detection theory, the existing HTD methods can be roughly divided into the following three categories.

Signal Detection Theory-Based Methods: Such methods are dominated by classical algorithms based on linear mixed models and statistical distribution hypothesis models. Among them, linear mixed model-based detection methods are represented by constrained energy minimization (CEM) [20] and orthogonal subspace projection (OSP) [3]. CEM is a linear filter, which allows the target-generated energy to pass through the filter while minimizing the unknown signal source energy. OSP projects the test spectrum into a subspace orthogonal with the background spectra to highlight the target. Based on CEM and OSP algorithms, some other detectors have been extended, such as the target-constrained interference minimized filter (TCIMF) [21], [22], [23]. Statistical distribution hypothesis-based models have been widely used in the past decades, for example, adaptive matched filter (AMF) [24], matched subspace detector (MSD) [25], and adaptive coherence/cosine estimator (ACE) [26]. Among them, ACE is considered one of the most effective detection methods, which treats hypothesis detection of hyperspectral targets as a binary problem and uses a generalized likelihood ratio test to obtain detection results.

Most of the above detection methods are based on prior statistical hypothesis models; however, these hypotheses usually fail to match the actual situation. Thus, although these detectors can detect most of the target pixels in HSI, this cannot meet the practical needs.

Machine Learning-Based Methods: In the past few years, machine learning methods have been successfully applied to HTD, such as kernel-based methods [27], [28], [29], [30], [31], [32], sparse representation-based methods [33], [34],

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[35], [36], [37], [38], [39], metric learning-based methods [40], [41], [42], [43], [44], and manifold learning-based methods [45], [46], [47], [48]. Kernel-based methods map the data to a high-dimensional potential kernel space, solving the linear indistinguishability problem in the original space, such as kernel-based constrained energy minimization (KCEM) [49]. Sparse representation-based methods use a small number of samples in the target and background dictionaries to linearly represent the HSI and obtain detection results based on the residual of the reconstructed sparse vectors, such as sparsity-based target detector (STD) [50]. Metric learning-based methods aim to measure the similarity between samples that bring samples of the same category closer together and samples of different categories farther apart, e.g., maximum margin metric learning (MMML)-based [51] target detection algorithm. Manifold learning-based methods use discriminative manifold embeddings to learn the subspace used to encode discriminative information. Recently, Zhu et al. [52] proposed a hyperspectral subpixel target detection method by learning single spectral abundance (denoted as LSSA).

Although the above methods achieve better detection results compared with traditional detection methods, they fail to mine the deep structural features and local interdependencies of the data.

Deep Learning-Based Methods: In recent years, deep learning models have attracted increasing attention by virtue of their ability to learn automatically and hierarchically from the data itself for deeper features [53]. It has been gradually applied to HSI classification [54], [55], [56], [57], [58]. There are relatively few efficient HTD methods based on deep learning. Li et al. [59] proposed a method called convolutional neural network (CNN)-based detection (CNND) utilizing labeled classification datasets to train the network, and the trained network is directly used for HTD. Obviously, it is unreasonable to equate HTD with hyperspectral classification. The reason is that there is often a lack of sufficient samples for network training in actual HTD. To overcome the few-training-sample issue, Zhang et al. [60] proposed a more reasonable version, an HTD framework using a deep network (HTD-Net). It utilized an autoencoder and linear strategy to obtain sufficient training samples. Zhu et al. [61] proposed a two-stream convolutional network-based target detector (TSCNTD) for HSIs, which utilized a synthetic method and sparse classification-based strategy to obtain a large number of training samples. Shi et al. [62] proposed a domain adaptive learning model for transferring knowledge learned from a large-sample to small-sample target domain. Compared with supervised networks, unsupervised networks do not need to consider the problem of insufficient training samples. For example, Li et al. [63] proposed a hyperspectral target method integrating an unconstrained linear mixture model and hierarchical denoising autoencoder (denoted as ULMMDL). Gao et al. [64] proposed a deep learning-based band selection hyperspectral target detection (DLBSTD) method. Qin et al. [65] proposed a proposal-weighted two-stage generative adversarial network (GAN) with attention mechanism method (denoted as PTGAN), which utilizes a GAN to estimate the spectral background distribution and combines attention mechanisms to perform feature interactions in the latent space, suppressing the background and highlighting the target. Furthermore, band selection-based [66], [67], robust signature-based [68], meta-learning-based [69], and transformer-based [70] HTD algorithms have exhibited competitive performances. They fully exploit and utilize the potential information of the data to obtain better detection performance and greater robustness than classical methods.

Despite some progress in deep learning-based HTD methods, the following challenging issues still need to be further explored to achieve better HTD performance.

1) It is well-known that depth models with a large number of parameters (usually tens of millions) require numerous labeled samples to avoid overfitting. However, in HTD applications, the available labeled samples are very limited, especially for prior targets. In order to successfully apply deep learning to HTD, the problem of insufficient training samples needs to be dealt with.
2) Deep network training requires not only a large number of labeled samples, but also representative samples to ensure that the network can learn effective data discrimination. In HTD applications, most of the HSIs are background pixels containing sparse targets and noise, and in order to make full use of this original background information, how to select pure background samples from the HSI is crucial to training the network's data discrimination ability.
3) The resource consumption and running speed of the model are critical to the HTD task. Therefore, model complexity (e.g., number of model parameters and computational complexity) should also be focused on when designing a more computationally efficient network model.

To address the three challenges of insufficient target samples, impure background samples, and complex models in deep learning-based HTD methods, this article proposes a lightweight convolutional neural network based on joint correlation distance constraints and density peak clustering (LCNN-CD) method for HTD. Specifically, the proposed method is divided into the following three steps in total to accomplish the HTD. First, the correlation coefficient and cosine distance between the prior target and the HSI are calculated, while the pixels with large coefficients and close distances are retained. Then, the retained pixel spectra are embedded in the prior target spectra by a certain ratio (random number from 0 to 1) to get a sufficient number of target samples. Second, the density peak clustering (DPC) algorithm is used to remove the sparse targets and noise in the HSI, followed by the combination of the prior target spectra to remove the target pixels that remain after the clustering processing, and the pure background samples are obtained. Finally, the training samples (including target and background samples) are input to a lightweight CNN designed for obtaining the final detection results. To verify the feasibility and effectiveness of the proposed method, we conduct comparison experiments with the state-of-the-art detection methods on three real hyperspectral datasets with different sensors. The experimental results show

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that the proposed method is superior in separating the target and background in HSI.

In summary, the main contributions of this article are as follows.

1) To address the problem of insufficient target samples in supervised networks, the proposed method uses the correlation and cosine distance between the prior target spectrum and the HSI as constraints to generate target samples. Compared with the strategies of expanding target samples with autoencoders or typical background pixels, the proposed method can effectively simulate the variability between the spectra of different target pixels while retaining the prior target spectrum information to the maximum extent.
2) To solve the problem of impure background samples in deep learning for HTD, the proposed method combines the prior target spectra with the DPC algorithm to extract pure background samples from the original HSI. The proposed method can cluster sparse targets and noise in HSI separately, which is beneficial for extracting pure background samples.
3) To accomplish the HTD task efficiently, the proposed method designed a lightweight CNN model with seven layers. It is well-known that fewer network layers and the number of neurons per layer directly affect the number of model parameters. The small number of layers of the proposed model is beneficial for improving computational efficiency. Compared with traditional detection methods, the proposed method requires no prior assumptions.

The rest of this article is organized as follows. Section II reviews the DPC algorithm and Pearson's correlation coefficient (PCC). Section III describes the proposed target detection method. Section IV demonstrates the superior performance of the method relative to other methods. Finally, Section V draws the conclusions and an outlook on future work.

II. Related Works

A. Pearson's Correlation Coefficient

The PCC [71] is translation invariant and scale invariant. It is based on the covariance matrix of the data to evaluate the strength of the relationship between two vectors. Let $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{M} \in \mathbb{R}^{M\times L}$ and $\mathbf{Y} = \{\mathbf{y}_j\}_{j=1}^{N} \in \mathbb{R}^{N\times L}$, where L presents the number of spectral bands, and M and N express the number of pixels. The PCC between $\mathbf{x}_i$ and $\mathbf{y}_j$ can be expressed as follows:

$$\operatorname{corr}\left(\mathbf{x}_i, \mathbf{y}_j\right) = \frac{\operatorname{cov}\left(\mathbf{x}_i, \mathbf{y}_j\right)}{\delta_{\mathbf{x}_i}\delta_{\mathbf{y}_j}} \tag{1}$$

where $cov(\cdot)$ is the covariance, and $\delta_{\mathbf{x}_i}$ and $\delta_{\mathbf{y}_j}$ denote the standard deviation of vector $\mathbf{x}_i$ and vector $\mathbf{y}_j$, respectively.

B. Density Peak Clustering

Clustering by fast search and finding of density peaks (DPC) [72] can automatically discover clustering centers and achieve efficient clustering of arbitrarily shaped data. This algorithm is based on two assumptions. First, the local density of the cluster center is assumed to be greater than the local density of the neighboring points around it. Second, the distance between the cluster center and any point with a higher local density is assumed to be relatively large. Under these two assumptions, the algorithm needs to calculate two quantities for each data point.

The first one is the local density of arbitrary data points. The local density ρ_i of data point i is defined as

$$\begin{aligned} \rho_{\mathrm{i}} &= \sum_j \chi\left(d_{ij} - d_c\right) \\ x &= d_{ij} - d_c \end{aligned} \tag{2}$$

where d_{ij} is the distance between point i and point j. d_c is a cutoff distance. x denotes the distance difference between d_{ij} and d_c, where $\chi(x) = 1$ if $x < 0$; otherwise, $\chi(x) = 0$. The local density ρ_i can be interpreted as the number of points whose distance from that point i is closer than the cutoff distance d_c.

The other one is the distance δ_i to the point with higher density. δ_i is measured by computing the minimum distance between the point i and any other point with higher density

$$\delta_i = \min_{j:\rho_j > \rho_i}\left(d_{ij}\right). \tag{3}$$

The point with the highest density usually takes $\delta_i = \max_j(d_{ij})$. Note that δ_i is much larger than the typical nearest neighbor distance only for points that are local or global maximum in the density. Thus, cluster centers are recognized as points for which the value of δ_i is anomalously large.

C. CNN-Based Methods

CNNs mimic the ability of visual cells to perceive objects locally by convolutional operations between adjacent nodes. This ability can extract local features of the spectrum, especially the subtle spectral differences between the target and the background. This facilitates the improvement of the detection performance of the model. It is well-known that depth models require a large number of labeled samples to train the network. However, for HTD, the number of target pixels is extremely limited, and the prior target spectrum that can be acquired is often only a few, which is not enough to train the network model.

To handle this problem of insufficient target samples, existing supervised network-based HTD algorithms usually utilize autoencoders or typical background pixels in the HSI to expand the target samples. However, the former requires training the autoencoder network to obtain the best network parameters for target samples generation (e.g., HTD-Net). The latter generates target samples by linear synthesis, e.g., the TSCNTD method, which embeds typical background pixel spectrum into the prior target spectrum in a certain proportion to obtain sufficient target samples. However, the quality of the target samples obtained in this way is limited by the abundance values of the embedded pixel spectra. In particular, the typical background spectra in HSI and the prior target spectra have large differences in shape. In addition, the TSCNTD method

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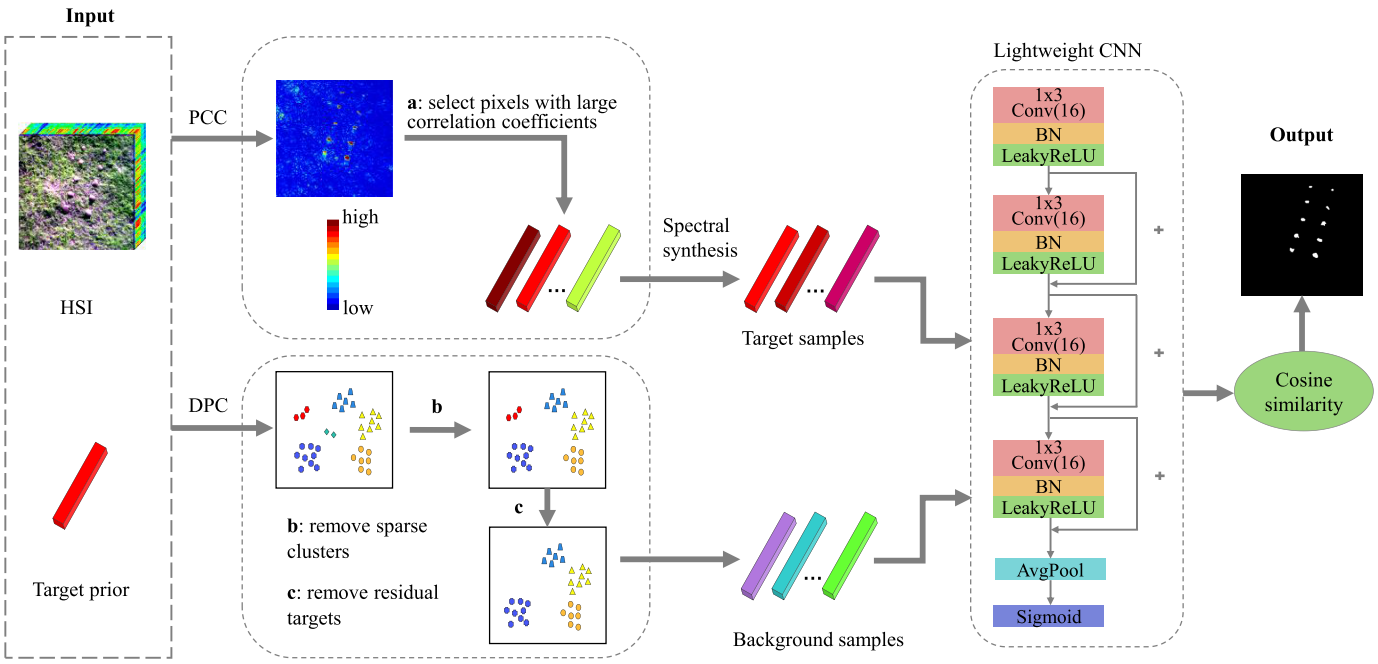

Fig. 1. Flowchart of the proposed LCNN-CD method. It is roughly divided into three steps. First, the target samples are generated. The PCC between the prior target spectrum and HSI is calculated, and pixels with large correlation are selected. Then, sufficient target samples are obtained using the spectral synthesis method. Second, pure background samples are extracted. The DPC is used to remove sparse targets and noises to obtain pure background clustering centers, and the pixels around the clustering centers are selected as background samples. Third, network training and target detection. The training samples (including target and background samples) are fed into the designed lightweight CNN, and their cosine similarity to the prior target is computed separately to obtain detection results.

uses K-means and sparse representation to obtain pure background samples. However, since the K-means method requires human intervention to set the number of categories, this is obviously unreasonable. At the same time, this method can misclassify the originally sparse target image elements into the background category, resulting in impure background samples. Therefore, pure background samples are also needed in supervised network models to ensure that the trained network has stronger robustness and data discrimination.

III. Proposed Method

Fig. 1 shows the overall flowchart of the proposed method. It is roughly divided into the following three steps.

1) First, the PCC between the prior target spectrum and the HSI are used as constraints to select pixels with high correlation. It is worth noting that there are phenomena of "same object with different spectrums" and "different objects with same spectrum" in the HSI. The pixels selected here may be the target pixels to be detected, or they may be background pixels very similar to the target. Then, the spectra of pixels satisfying the constraints are embedded into the prior target spectra in a certain proportion (i.e., spectral synthesis method) to obtain sufficient target samples.
2) Second, sparse targets and noise are removed using DPC. Specifically, since the DPC can automatically discover cluster centers, efficient clustering of arbitrarily shaped data is achieved. After the original HSI goes through DPC, sparse targets and noise are clustered into separate classes. At that time, we remove the sparse clustering terms, which can initially remove the sparse targets and noise. However, considering that the number of target pixels to be detected is relatively large, the target pixels to be detected cannot be completely eliminated by the previous step. Then, we use the prior target spectrum to judge the similarity with all the remaining clustering centers and further eliminate the residual target pixels to obtain the pure background clustering centers. Finally, we select the pixels around these background clustering centers in a certain proportion to obtain pure background samples. Notably, the sparse background classes may be eliminated during the initial rejection of sparse targets and noise, but this does not affect the detection performance of the proposed method. Because, in this case, it can be seen that the selected pure background samples are mainly from the main background categories in HSI.
3) Finally, we feed the target and background samples into the designed lightweight CNN to obtain their high-level features. Their cosine similarity to the prior target spectrum is used as the final detection result.

A. Target Samples Generation

To address the problem of insufficient target samples in supervised networks, we use PCC distance as constraints to select pixels for the target samples generation, as shown in Fig. 1. In order to ensure that the expanded target samples can simulate the subtle differences between the target spectrums, and to maximize the expanded target samples to retain the original prior target spectral information, we conducted an

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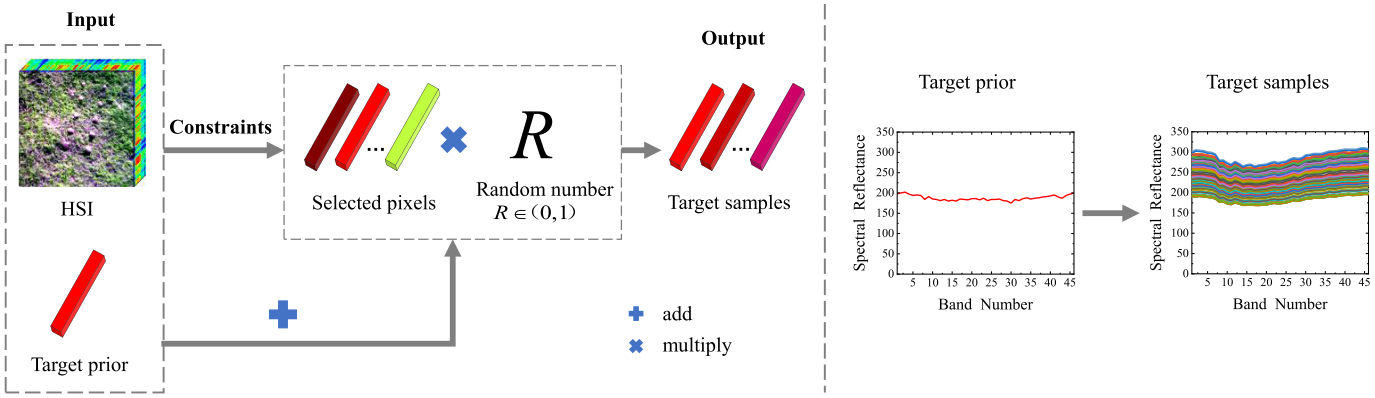

Fig. 2. Flowchart of generating target samples. The constraint is the Pearson correlation distance between the prior target spectrum and the HSI. Pixels with high correlation are selected for generating target samples. The abundance value R represents a set of random numbers between 0 and 1. Expressed in an expression: prior target + selected pixel $\times R$ = target samples.

experimental analysis on the number of selected pixels. The results show that on different experimental datasets, taking the first three pixels with the highest correlation can make the quality of the generated target samples better.

Due to the limitation of the spatial resolution of HSI, the spectral information of a single pixel is usually acquired as a mixture of the spectra of multiple materials. Meanwhile, the spectra of different pixels in the same material exhibit variability, i.e., the spectra of similar materials are not identical. Therefore, we use the spectral synthesis method to generate sufficient target samples. Specifically, the spectra of selected pixels are implanted into the prior target spectrum in a certain proportion to realize the target samples generation, as shown in Fig. 2.

There are two advantages of this strategy. On the one hand, it avoids the situation in which there is a large spectrum shape difference between the image pixels used to expand the target samples and the prior target spectrum. On the other hand, it effectively simulates the differences between the target spectrums while retaining the prior target spectral information.

Due to the variability of the spectrum of features in HSI, the spectra of different materials are superimposed, and the difference in spectral characteristics of similar materials varies greatly. The target sample generation is achieved by mixing the selected pixels (maybe target or hard background pixels) that satisfy the constraints with the prior target spectrum, considering both the mixing of different spectrums among target pixels and the mixing of target pixels with background pixels.

B. Pure Background Samples Extraction

To overcome the problem of impure background samples in supervised networks, we combine the prior target spectrum and the original HSI to extract pure background samples using the DPC algorithm. Fig. 3 shows the process of extracting pure background samples from HSI.

Compared with other clustering methods, the DPC can automatically discover clustering centers and achieve efficient clustering of arbitrarily shaped data. Therefore, its application to HTD can achieve separate clustering of sparse targets and noise. This facilitates the rejection of targets and noise contained in the HSI, thus obtaining relatively pure background pixels.

However, there are cases where the number of target pixels in the HSI is relatively large, and then, all the target pixels cannot be eliminated by simply sorting the local density between pixels. This will lead to some target pixels still existing in the selected background samples, i.e., the background samples are impure. To further remove the residual target pixels to obtain a purer clustering center of the background pixels. We combine the prior target spectrum with the clustering center of the sparse targets and noise removed by the cosine similarity measure.

Finally, we select the pixels around each cluster center as background samples based on the ratio of the number of pixels in each cluster to the number of pure background pixels. This strategy obtains pure background samples and makes full use of the spectral information of the original HSI.

C. Lightweight CNN

1) Network Architecture: The lightweight convolutional network architecture proposed in this article is shown in Fig. 1. The network consists of four convolutional layers, a mean pooling layer, and a sigmoid output layer. There are only 16 neurons per convolutional layer, which greatly reduces the number of network parameters and helps to improve the computational efficiency.

The first four layers are 1-D convolutional layers for spectral feature information extraction, where the output of each layer is superimposed on the input of the next layer, which aims to extract new potential features while minimizing the loss of the original spectral information. The convolutional kernel size of all convolutional layers is 1×3. The stride size is set as 1, and the number of neurons is 16. To avoid exploding/vanishing gradients and speed up training, each convolutional layer is followed by batch normalization (BN) and leaky linear rectification (LeakyReLU) function activation function. Since the convolution operation can only extract local features of the spectrum, the fifth layer is set as the average pooling (AvgPool) layer, obtaining the global features of the spectrum. Since the output value of the sigmoid function is in

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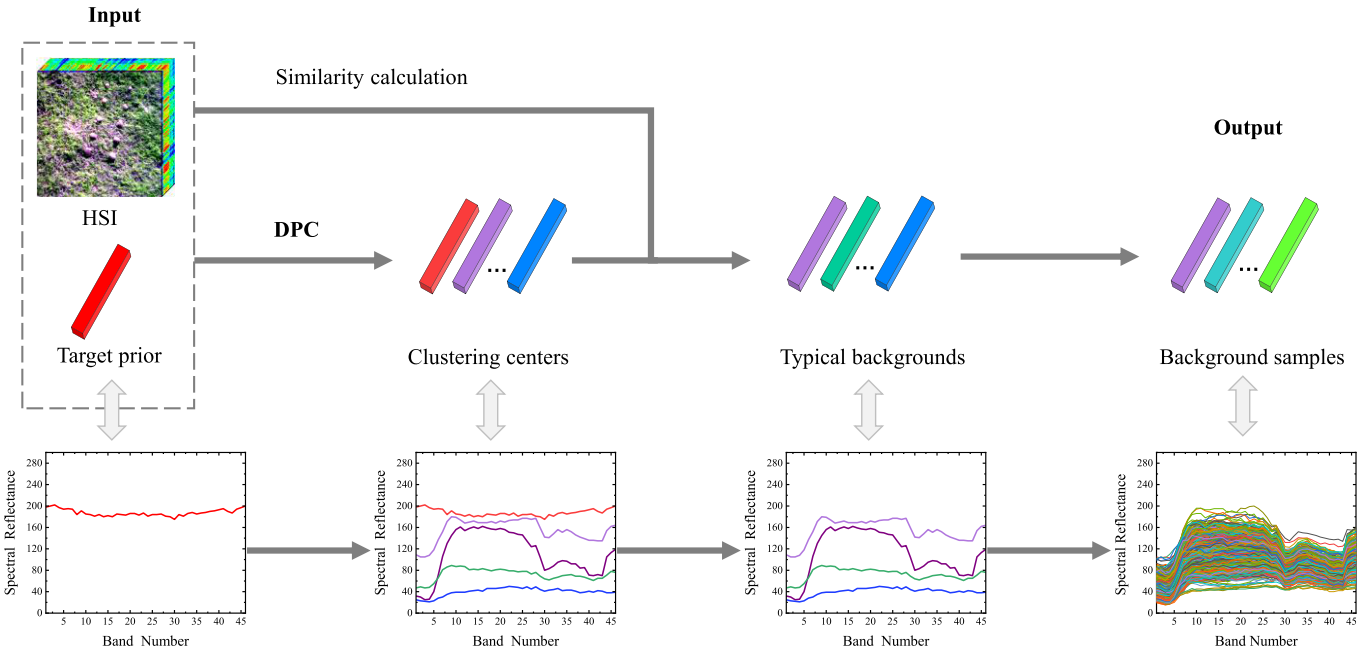

Fig. 3. Flowchart for selecting pure background samples.

the range of 0–1, it can be represented as the normalization of the output value, which is helpful to improve the accuracy of the subsequent similarity calculation.

Therefore, we use it as the network output layer. Except for the sigmoid layer, the padding of the remaining five layers is 1. Note that the input of this network is the sample pairs of the prior target and the target/background samples (or the prior target and the pixels to be tested), and the output is the high-level feature extracted by the network. $\{\mathbf{z}_i\}_{i=1}^{N} \in \mathbb{R}^{N\times L}$ denotes the target/background sample spectrum, where L presents the number of spectral bands and N expresses the number of samples. $\mathbf{z}_0$ in $\mathbb{R}^L$ denotes the prior target spectrum. For the input sample pairs $(\mathbf{z}_0, \mathbf{z}_i)$, the extracted features of the network are represented as $\{f(\mathbf{z}_0), f(\mathbf{z}_i)\}$. We use the contrast loss function [73] to guide the model training with the following equation:

$$\begin{aligned}\text{Loss}(\mathbf{z}_0, \mathbf{z}_i, y) = {} & y||f(\mathbf{z}_0) - f(\mathbf{z}_i)||_2^2 \\ & + (1-y)\left\{\max\left(0, m-||f(\mathbf{z}_0)-f(\mathbf{z}_i)||_2^2\right)\right\}^2\end{aligned} \tag{4}$$

where y is the sample pair label, and $f(\cdot)$ denotes the network output value. m is a hyperparameter that indicates the degree of distance between control negative sample pairs (target and background).

2) Training and Detection: After obtaining the target and background samples, they are formed into positive and negative pixel pairs with the prior target spectrum, respectively. Then, the positive and negative sample pairs are input into the designed network for training. Specifically, the prior target and the target samples form positive sample pairs with a label of 1, representing that the training samples are similar to the prior target. The prior target and the background samples form negative sample pairs with a label of 0, which means the training samples are not similar to the prior target. The network learns the similarity discrimination ability of the data by learning the similarity between pixel pairs. Since the performance of CNN is affected by the distribution of training samples [74], in this article, we use the same number of positive and negative sample pairs to make a balanced distribution of training samples.

Consider the given hyperspectral cube is $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{M} \in \mathbb{R}^{M\times L}$, where L presents the number of spectral bands, M expresses the number of pixels, and $\mathbf{x}_i$ represents the pixel in HSI. Finally, the cosine similarity between network outputs is calculated to obtain the final detection results. Considering that sometimes there are multiple prior target spectrums, in order to reduce the variability among the prior target spectrums, we use the given spectral mean value as a prior information input to the network. The final output of LCNN-CD is

$$D_{\text{LCNN-CD}}(\mathbf{x}_i) = \frac{f(\mathbf{z}_0)\cdot f(\mathbf{x}_i)}{||f(\mathbf{z}_0)|| \times ||f(\mathbf{x}_i)||} \tag{5}$$

where $||\cdot||$ denotes the mode of the vector. If $D_{\text{LCNN-CD}}(\mathbf{x}_i)$ is larger than the threshold, $\mathbf{x}_i$ can be classified as a target pixel; otherwise, $\mathbf{x}_i$ is a background pixel.

IV. Experiments

In this section, we conduct experiments on real hyperspectral datasets from three different sensors and analyze them in comparison with classical detection methods to demonstrate the effectiveness and robustness of the proposed method.

A. Experimental Details

1) Data Description:

Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) San Diego Dataset: The first dataset is the airport data of San Diego, CA, USA, acquired by the AVIRIS sensor, which contains the aircraft, housetop, and airport runway, as shown

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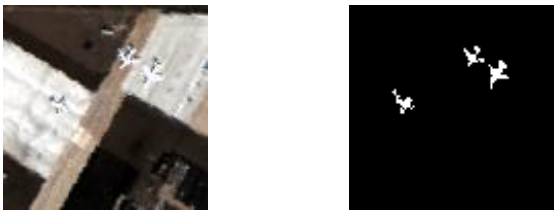
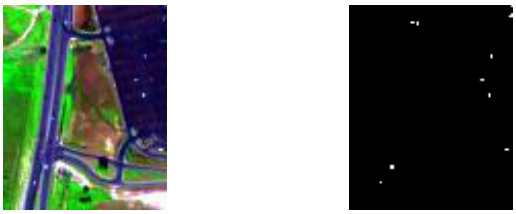


(a) (b)

Fig. 4. AVIRIS San Diego dataset: (a) image scene and (b) target locations.



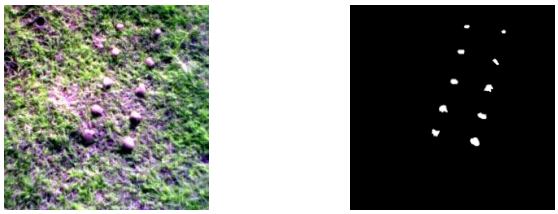
(a) (b)

Fig. 5. HYDICE Urban dataset: (a) image scene and (b) target locations.

in Fig. 4. The experimental data image size is 100 × 100, the spectral resolution is 10 nm, the spatial resolution is 3.5 m, and the band range is 370–2510 nm. The water vapor absorption bands in the wavelength range of 1350–1420 and 1810–1940 nm are removed, including 1–6, 33–35, 97, 107–113, 153–166, and 221–224 bands, reserving 189 usable bands. The aircraft was selected as the target of interest in the experiment, and it accounts for 0.0134 of HSI.

Hyperspectral Digital Imagery Collection Experiment (HYDICE) Urban Dataset: The second dataset is the hyperspectral data collected by the HYDICE sensor for the urban area near Fort Hood, TN, USA. The image size of the experimental data is 80 × 100, with a spectral resolution of 10 nm and a spatial resolution of 2 m. There are 210 bands covering the wavelength range of 400–2500 nm. Some bands with no value or low signal-to-noise ratio due to strip noise and water absorption phenomenon are removed, including bands 1–4, 87, 101–111, 136–153, 198–210, and 162 usable bands are retained. Vehicles, highways, parking lots, houses, trails, woods, and soil are included in this data. The vehicle is selected as the target of interest in the experiment, and it accounts for 0.002625 of the HSI. It is noted that there are many vehicles on roads, parking lots, and residential areas. These complex environmental factors increase the difficulty of vehicle detection because these vehicles have different materials, as well as the road materials vary greatly (including asphalt roads and concrete roads), etc. Fig. 5 shows the false color image scene and the location of the target for this dataset.

Nuance CRI Grass-Stone Dataset: The third dataset is a real HSI acquired by Nuance's CRI hyperspectral imager, a handheld imager with a spectral resolution of about 10 nm in 46 bands and a spectral range of 650–1100 nm. The entire image is 400 × 400 pixels and contains human-arranged ten stone targets of interest and several background features (e.g., bare soil and grass). The stone is selected as the target of interest in the experiment, and it accounts for 0.007825 of the HSI. Due to different sizes and materials of the target stones, as well as the solar altitude angle which makes the stones



(a) (b)

Fig. 6. Nuance CRI Grass-Stone dataset: (a) image scene and (b) target locations.

produce some shadows, it can cause many difficulties in stone detection. Fig. 6 shows the false color image scene and the location of the target in this dataset.

2) Evaluation Criterion: We use the following three quantitative metrics to evaluate the detection performance of each detection methods: receiver operating characteristic curve (ROC curve), area under ROC curve (AUC value), and target-background separability map.

The ROC curve [75] is a widely applied tool for evaluating detectors. It is a curve composed of a set of points with the horizontal coordinate of the false alarm rate and the vertical coordinate of the detection rate. When the ROC curve is closer to the upper right, it indicates better the detection performance of the detector. In general, we focus on the detection rate when the false alarm rate is less than 0.001.

The AUC value represents the area under the ROC curve. It can effectively deal with the problem that it is difficult to judge the detection performance of each detection method when the ROC curve crosses. The value of AUC is between 0 and 1. Among them, the larger the AUC value, the better the detection performance.

The target-background separability map is a box figure to represent the separability of the target and background classes. Its vertical coordinate is normalized to the detection statistical range between 0 and 1. In the target-background separability map, the green box represents the range of the target class and the orange box represents the range of the background class. The square box represents the portion of the image element between 10% and 90%. In the target background separation diagram, there are three points that require medium attention. First, the higher the position of the green box in the figure, the more effective the detection method is in highlighting the target. Second, the lower the position of the orange box in the figure, the better the detection method suppresses the background effect. Third, when the interval between the green and orange boxes is larger, it means that the detector has better separability between the target and the background. When the interval between the green box and the orange box is larger, it means that the detector has better separability between the target and the background.

3) Experimental Settings: In order to demonstrate the detection performance and robustness of the proposed LCNN-CD method, six detection methods are selected for comparison and analysis with the proposed algorithm. The comparison methods include ACE [26] and CEM [20] based on signal detection theory, STD [50], MMML [51], and LSSA [52] based on machine learning method, and HTD-Net [60] and

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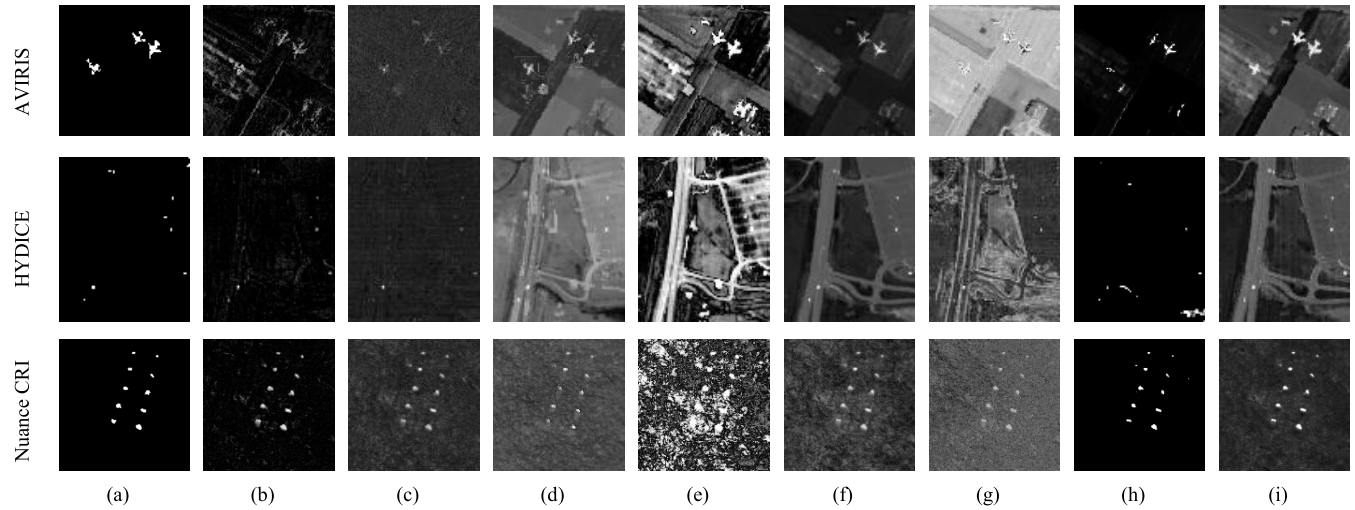

Fig. 7. Detection maps of the compared methods: (a) mask; (b) ACE; (c) CEM; (d) STD; (e) MMML; (f) LSSA; (g) HTD-Net; (h) TSCNTD; and (i) LCNN-CD.

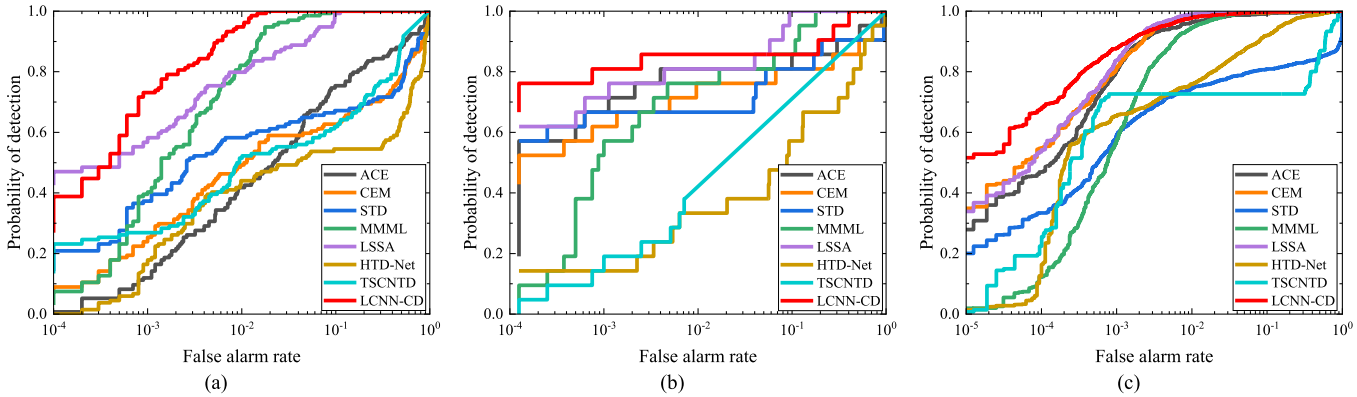

Fig. 8. ROC curves for all detectors on the three datasets: (a) AVIRIS San Diego; (b) HYDICE Urban; and (c) Nuance CRI Grass-Stone.

TSCNTD [61] based on deep learning method. To demonstrate the universality of the proposed method, only one prior target spectrum is used in this article because the prior target spectrum is extremely limited in practical applications. At the same time, to ensure the fairness of the experiment, all methods use the same prior target spectrum for the same dataset. In addition, for detection methods involving parameter settings, we take the set of parameters with the best detection results to obtain the target detection results.

According to [50], the STD method requires setting a double window size to simulate the local background, and the window size is set by the target size of interest in the dataset. In this article, the sizes of the outer and inner windows of the AVIRIS dataset are 19 $\times$ 19 and 17 $\times$ 17, respectively. The sizes of the outer and inner windows of the HYDICE dataset are 13 $\times$ 13 and 5 $\times$ 5, respectively. And the sizes of the outer and inner windows of the Nuance CRI dataset are 23 $\times$ 23 and 21 $\times$ 21, respectively. In addition, the corresponding sparsity of these three datasets is set to 3, 2, and 9, respectively. For the MMML method, in addition to the target prior information, additional prior background information is required, and we follow the recommendations in the reference to select the background prior. That is, a number of prior background pixels selected on the AVIRIS, HYDICE, and Nuance CRI datasets are 20, 17, and 20 in turn, respectively.

The deep learning-based detection methods need to be tuned by the network parameters learning rate, epoch number, and batch size to obtain an optimal set of parameters to achieve good detection results on different datasets. In this article, the three sets of parameters, learning rate, epoch number, and batch size, are set to 1e-2, 50, and 1024 for the HTD-Net, and 1e-3, 50, and 256 for the TSCNTD. For LCNN-CD, they are set to 1e-3, 50, and 64. For the deep model detection method, it is necessary to input target samples and background samples to train the network to obtain the discriminative ability of the target and the background. For the sake of fairness, the number of background pixels selected from the HSI by these three methods is kept the same, which is set to 2000 in this article.

B. Detection Performance

Fig. 7 visualizes the detection results of LCNN-CD and other comparison methods. CEM and STD can detect most of the target pixels, but some of the detected target pixels are not obvious. Compared with the proposed LCNN-CD method, ACE and TSCNTD methods are more effective in suppressing

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TABLE I
AUC VALUES' COMPARISON FOR THREE DATASETS

Methods	AVIRIS	HYDICE	Nuance CRI
ACE [27]	0.8512	0.9103	0.9879
CEM [20]	0.7692	0.8646	0.9899
STD [51]	0.7760	0.8824	0.8378
MMML [52]	0.9806	0.9733	0.9892
LSSA [53]	0.9755	0.9632	0.9914
HTD-Net [61]	0.6594	0.7796	0.9650
TSCNTD [62]	0.8200	0.6853	0.8493
LCNN-CD (ours)	**0.9951**	**0.9867**	**0.9985**

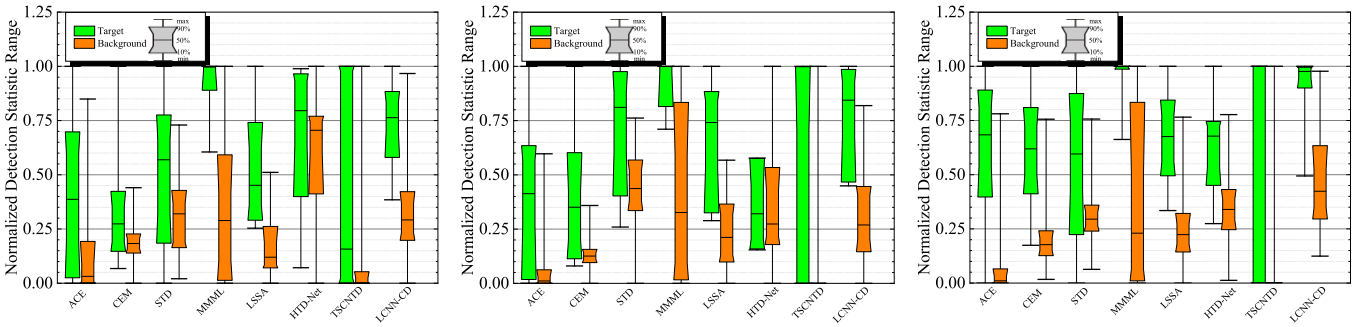

Fig. 9. Maps of target background separation for all detection methods on the three datasets: (a) AVIRIS San Diego; (b) HYDICE Urban; and (c) Nuance CRI Grass-Stone.

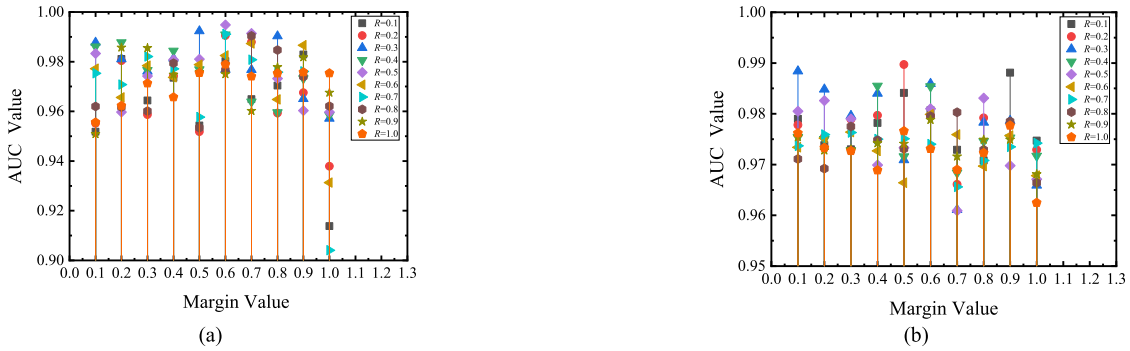

Fig. 10. Parametric analysis chart: (a) AVIRIS San Diego and (b) HYDICE Urban.

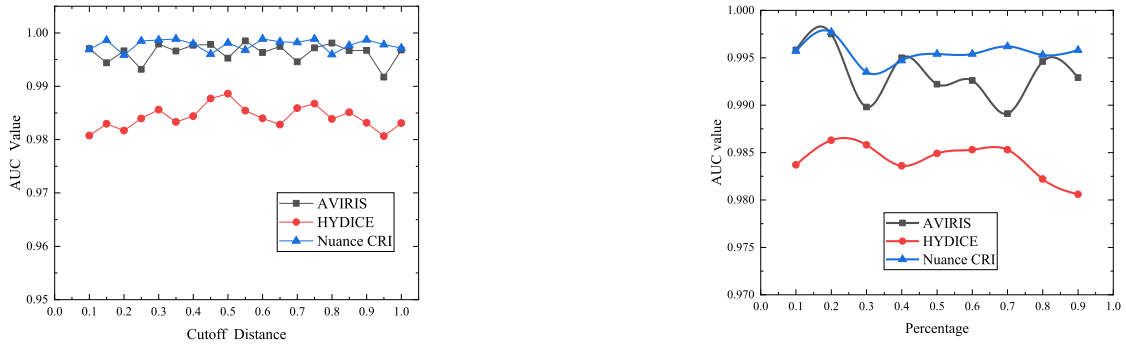

Fig. 11. Analysis of the cutoff distance.

Fig. 12. Analysis of removal percentage.

the background, but their detection results on the three datasets are unstable. Specifically, although ACE and TSCNTD have better detection results on the Nuance CRI dataset, some target pixels are missed on the other two datasets, and some pixels are incorrectly detected as targets in the TSCNTD method. Overall, the detection rate of the proposed LCNN-CD is more stable on the three datasets, and most of the target pixels can

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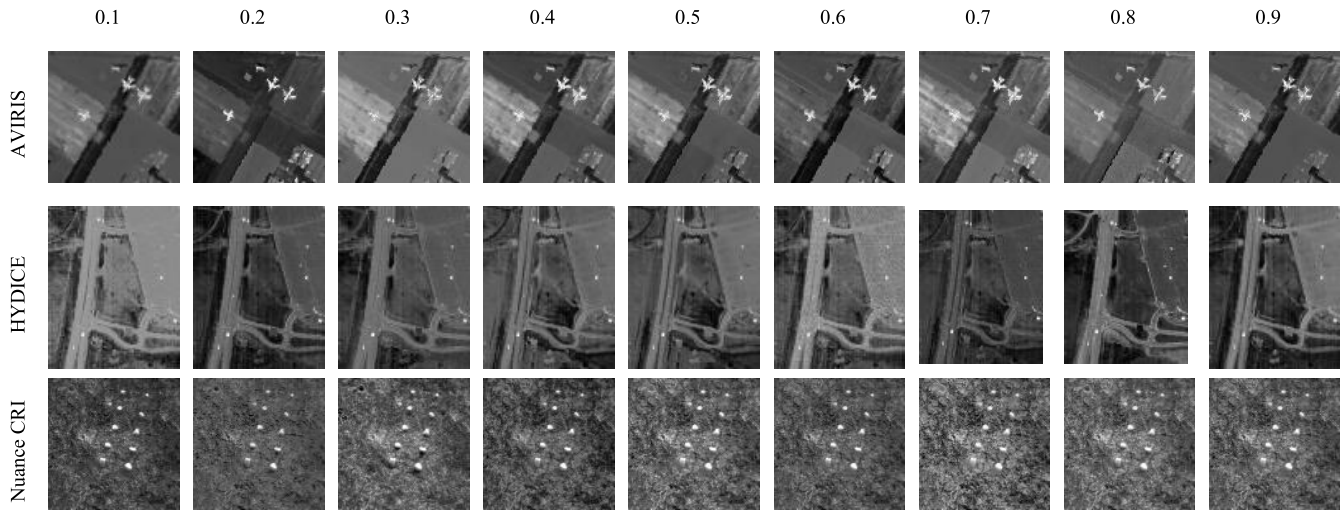

Fig. 13. Detection maps of different percentages.

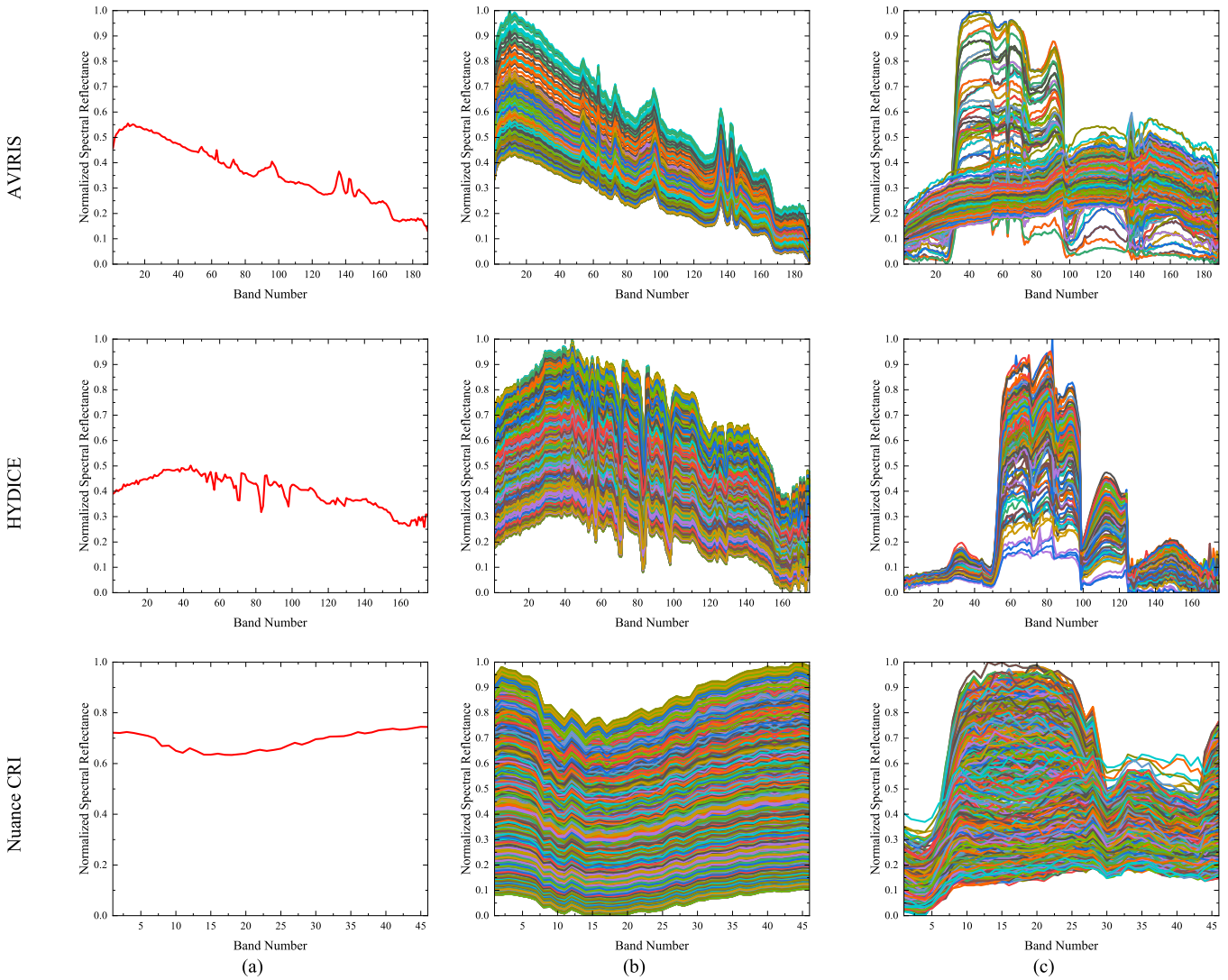

Fig. 14. Visualization of sample spectra curves on three real HSIs: (a) prior target; (b) target samples; and (c) background samples.

be detected correctly, while the target pixel highlighting effect is more obvious.

Fig. 8 shows the ROC curves, where the red curve represents the proposed detector. As can be seen from Fig. 8, the

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TABLE II
COMPUTATIONAL EFFICIENCY OF ALL DETECTORS ON THREE DATASETS (UNIT: SECOND)

Methods	AVIRIS	HYDICE	Nuance CRI
ACE [27]	1.178	1.012	2.027
CEM [20]	0.223	0.210	0.609
STD [51]	8.544	4.184	115.631
MMML [52]	7.798	6.774	8.591
LSSA [53]	2.036	1.601	30.562
HTD-Net [61]	344.543	322.331	529.472
TSCNTD [62]	40.052	36.183	194.364
LCNN-CD (ours)	15.543	14.714	68.512

horizontal coordinate is the false alarm rate and the vertical coordinate is the detection rate. When the false alarm rate is less than 0.001, the red curve is higher than the other color curves on all three datasets, which indicates that the proposed LCNN-CD method has a more stable detection effect. In addition, it can be seen that the ROC curves of the other two deep model detectors on the three sets of data are relatively low. All experimental datasets in this article use only one prior target spectrum, while the two detection methods in the original literature use multiple prior target spectra; this may be the reason for the degraded detection performance of these two detectors.

From the comparison of the AUC values of all methods in Table I, the AUC values of MMML and LCNN-CD are relatively stable and reach above 0.9 on different datasets. And the LCNN-CD method has a larger AUC value, which indicates that it has a better detection performance. In general, the AUC values of the two depth model detectors compared are significantly lower than those of the traditional and machine learning detectors, which may be caused by only one prior target spectrum given in the experiment.

In addition, to test the effectiveness of the proposed method in separating the target from the background, we show the target-background separability of each method on the three datasets, as shown in Fig. 9. It can be clearly seen from Fig. 9 that the interval between the green box (target class) and the orange box (background class) corresponding to the proposed LCNN-CD method is more obvious on all three datasets. In addition, the green box (target class) is at a higher position in the LCNN-CD method, which indicates that the proposed method can also highlight the target better.

Table II provides the calculation cost (time unit: second) of the proposed method and the comparison method on the three sets of datasets. Among them, for the deep model detection method, the training time and testing time are added together. As can be seen from Table II, traditional detection methods (ACE and CEM) and machine learning methods (STD and MMML) generally take less time than deep learning-based methods (HTD-Net, TSCNTD, and LCNN-CD). In addition, as the experimental dataset increases, the time spent by each time than the other two deep learning model methods. Compared to the CNN with 16 convolutional layers and 30 neurons in each layer used by HTD-Net and TSCNTD methods, the proposed lightweight network uses only four convolutional layers with only 16 neurons in each layer, which greatly reduces the amount of network parameters. It is beneficial to improve the computational efficiency while ensuring the network detection accuracy.

TABLE III
MODEL COMPUTATION AND INFERENCE SPEED OF CONTRASTED DEPTH METHODS

Methods	HTD-Net	TSCNTD	LCNN-CD
Total params/(PC)	8581	6690	912
Totoal memory/(MB)	0.08	0.08	0.08
Total Madd/(KMAdd)	6.63	8.55	0
Total Flops/(KFlops)	6.6	7.68	6.05
Total MemR+W/(KB)	51.57	56.84	47.25

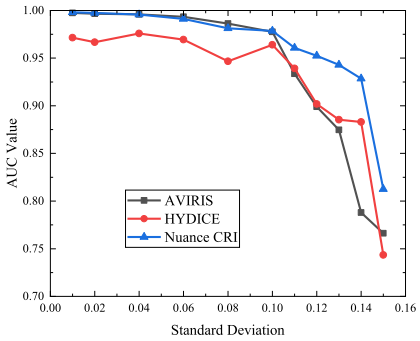

Fig. 15. Detection accuracy of LCNN-CD under different levels of Gaussian noise interference.

Table III shows the speed of model computation and inference for the depth methods compared in this article. It can be seen that the computational efficiency of the proposed LCNN-CD method is more advantageous.

C. Parameter Analysis

Four important parameters of the proposed HTD method are parametrically analyzed here. The first parameter is R, the abundance value used to implant the image element spectra that satisfy the constraint into the prior target spectrum. The second parameter is m, which controls the degree of target-background separation in the contrast loss function. These two parameters take values ranging from 0 to 1. The results of the parametric analysis of this parameter are not visualized because of its insignificant effect on the Nuance CRI Grass-Stone dataset. Only the effect of this parameter on the detection performance of the AVIRIS San Diego and HYDICE Urban datasets is shown here.

Fig. 10 shows the results of the influence of these two parameters on the AUC of the detection method. In Fig. 10, the abscissa indicates the margin value, the ordinate indicates the AUC value, and different symbols indicate that the parameter R takes different values. It can be seen from Fig. 10 that when the margin is 0.6, no matter how much the parameter R is; the corresponding AUC value is higher than other margin values. On the whole, regardless of the value of margin and R, the corresponding AUC is above 0.9.

The third parameter is the cutoff distance d_c in DPC, which can be regarded as the radius size of the current cluster. In this

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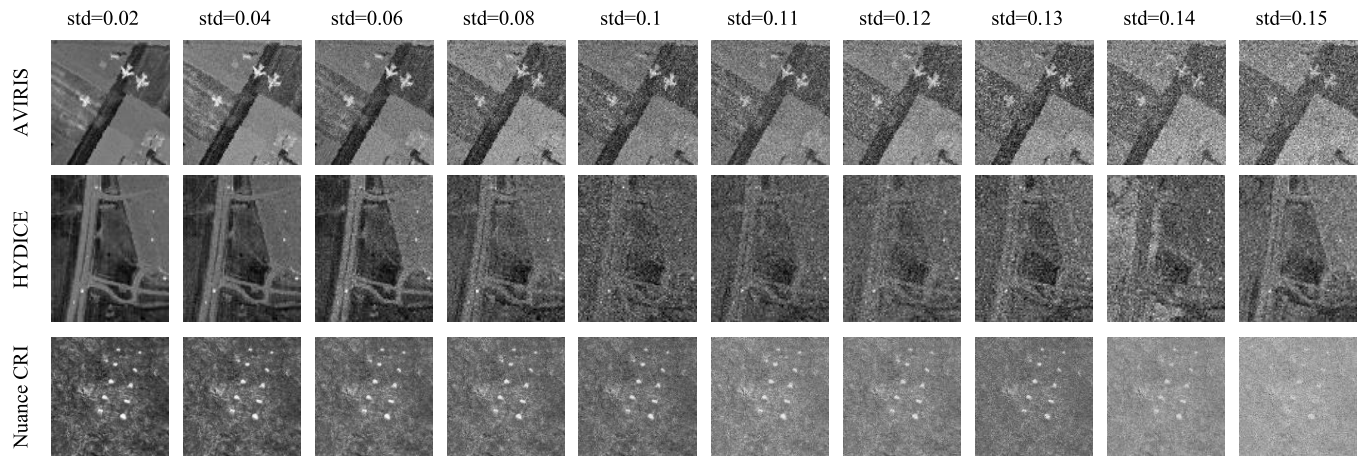

Fig. 16. Detection maps of LCNN-CD with different levels of Gaussian noise interference.

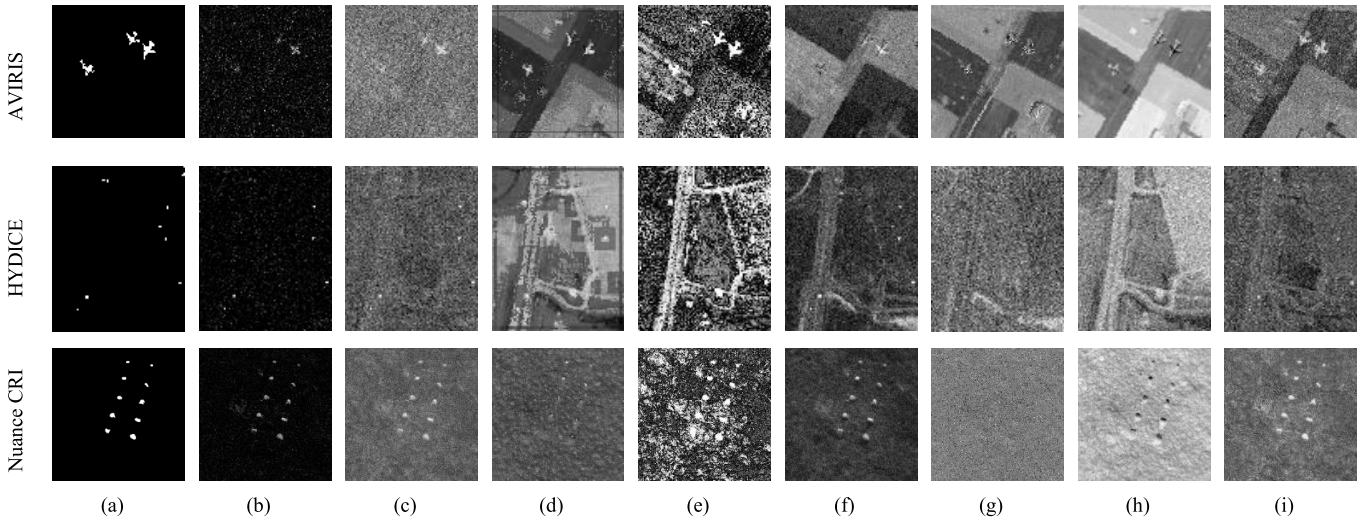

Fig. 17. Detection maps of the comparison methods after adding Gaussian noise: (a) mask; (b) ACE; (c) CEM; (d) STD; (e) MMML; (f) LSSA; (g) HTD-Net; (h) TSCNTD; and (i) LCNN-CD.

article, the distance between samples is measured using cosine similarity. Therefore, the distance values are in the range of 0–1. We took d_c values at 0.05 intervals for the analysis and plotted the changes in the corresponding AUC values. As can be seen from Fig. 11, the AUC values are relatively stable as d_c changes over the three datasets. In our experiments, we set this parameter to 0.5.

The fourth parameter is the percentage of residual targets to be removed when extracting the pure background. The relationship between the selected percentage and its detection accuracy (AUC value) is illustrated in Fig. 12. Considering that the residual targets are small in number, we control the percentage between 0 and 1. As can be seen from Fig. 12, when the percentage is 0.2, the AUC value is relatively high on the three datasets. In addition, we also show the detection results corresponding to the percentages, as shown in Fig. 13.

D. Sample Visualization

To visually demonstrate the quality of the samples acquired by the LCNN-CD method, we visualized the spectral shape of the samples. Fig. 14 shows the visualization results of target samples and selected background samples generated by the LCNN-CD method on three real HSIs datasets. Fig. 14(b) shows that the generated target samples greatly retain the prior target spectral information, and there are some spectral differences between the target samples. Fig. 14(c) shows that the background and target samples have extremely different shapes, indicating that the selected background samples are pure.

E. Antinoise Analysis

The noise in hyperspectral imagery can generally be categorized into two classes: random noise and fixed-pattern noise. Fixed-pattern noise like striping is generated during the calibration process. Random noise is randomly generated by environmental influences. One widely used random noise model in HSI is the additive model, which is assumed to be white, Gaussian, and independent from the signal. In this work, we added Gaussian noise with a mean value of 0 and different standard deviations as noise interference to HSIs to

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test the anti-interference capability of the proposed method. Fig. 15 shows the detection accuracy at different noise levels. When the standard deviation is within 0.1 or less, the detection accuracy of LCNN-CD is relatively stable. When the standard deviation is greater than 0.13, the AUC value decreases sharply. Fig. 16 shows the detection maps of LCNN-CD with different levels of Gaussian noise interference under three sets of real HSIs. The detection results shown correspond to Fig. 15. Fig. 17 shows the detection results of the compared methods after adding Gaussian noise with a mean value of 0 and a standard deviation of 0.1 to all datasets. It can be seen that the detection results of the proposed method are relatively stable for each dataset.

V. Conclusion

In this article, we propose an effective LCNN-CD method for HTD. To address the problem of insufficient target samples in deep learning, this method uses the correlation distance constraint between the prior target and the HSI to generate target samples. For the problem of impure background samples, the proposed method utilizes DPC to extract pure background samples. Finally, a simple lightweight CNN is used to complete the target detection task. The experimental results and detailed analysis of three real-world HSIs demonstrate the effectiveness of the proposed method in comparison with other state-of-the-art methods. Meanwhile, the LCNN-CD method is still insufficient in suppressing the background. After careful analysis, it may be that the model used does not fully exploit the deep feature differences of the samples. This will be improved in the future to achieve better background suppression.

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